

Electrically operated food preparation appliances — Methods for measuring the performance

The European Standard EN 60619:1993, with the incorporation of A1:1995 and A2:2004, has the status of a British Standard

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Cooperating organizations

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National foreword

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This British Standard supersedes BS 3999-3:1967 which is withdrawn.

The start and finish of text introduced or altered by amendment is indicated in the text by tags **A1** **A1**. Tags indicating changes to IEC text carry the number of the amendment. For example, text altered by IEC amendment 1 is indicated by **A1** **A1**.

NOTE In 3.1.12 the term should be “french fry chipping”.

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Summary of pages

This document comprises a front cover, an inside front cover, pages i and ii, the EN title page, pages 2 to 26, an inside back cover and a back cover.

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English version

Electrically operated food preparation appliances — Methods for measuring the performance (includes amendments A1:1995 and A2:2004)

(IEC 619:1993 + A1:1995 + A2:2004)

Appareils électriques pour la préparation de la
nourriture —
Méthodes de mesure de l'optitude à la fonction
(inclut les amendements A1:1995 et A2:2004)
(CEI 619:1993 + A1:1995 + A2:2004)

Elektrische Kuchenmaschinen —
Prüfverfahren zur Bestimmung der
Gebrauchseigenschaften
(enthält Änderungen A1:1995 und A2:2004)
(IEC 619:1993 + A1:1995 + A2:2004)

This European Standard was approved by CENELEC on 1992-12-09. Amendment A1 was approved by CENELEC on 1995-09-20; amendment A2 was approved by CENELEC on 1 September 2004. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

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CENELEC

European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

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Foreword

The text of document 59G(CO)25, as prepared by sub-committee 59G: Small kitchen appliances, of IEC technical committee 59: Performance of household electrical appliances, was submitted to the IEC-CENELEC parallel vote in March 1992.

The reference document was approved by CENELEC as EN 60619 on 9 December 1992.

The following dates were fixed:

- latest date of publication of an identical national standard (dop) 1994-02-01
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Foreword to amendment A1

The text of document 59G/49/DIS, future amendment 1 to IEC 619:1993, prepared by SC 59G, Small kitchen appliances, of IEC TC 59, Performance of household electrical appliances, was submitted to the IEC-CENELEC parallel vote and was approved by CENELEC as amendment A1 to EN 60619:1993 on 1995-09-20.

The following dates were fixed:

- latest date by which the amendment has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 1996-07-01
- latest date by which the national standards conflicting with the amendment have to be withdrawn (dow) 1996-07-01

Foreword to amendment A2

The text of document 59G/156/FDIS, future amendment 2 to IEC 60619:1993, prepared by SC 59L, Small household appliances, of IEC TC 59, Performance of household electrical appliances, was submitted to the IEC-CENELEC parallel vote and was approved by CENELEC as amendment A2 to EN 60619:1993 on 2004-09-01.

The following dates were fixed:

- latest date by which the amendment has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2005-06-01
- latest date by which the national standards conflicting with the amendment have to be withdrawn (dow) 2007-09-01

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1 Scope and object

1.1 Scope

This International Standard applies to electrically operated food preparation appliances for household use.

The purpose of this standard is to state and define test methods of measuring the functions that can be carried out by means of household electrical food preparation appliances, which are of interest to the user and to give some guidelines for the evaluation of test results.

Taking into account the lower grade of accuracy and repeatability, due to variations in time and origin of test materials and ingredients and to the influence of the subjective judgement of test operators, the described test methods may be applied more reliably for comparative testing of a number of appliances at approximately the same time, in the same laboratory, by the same operator and with the same utensils, rather than for testing of single appliances in different laboratories.

As there is no definition of a given type or size of oven, and as a number of the tests involve baking of the final product in order to make a determination of volume, a variation in results can be expected between ovens used. All comparative tests should be under taken in the same oven.

1.2 Aspects excluded from the scope

This standard is not concerned with safety.

It does not apply to appliances designed exclusively for commercial or industrial use.

Attention is drawn to the fact that sometimes the same result may be obtained using different functions.

2 Normative references

Void.

NOTE Other clauses which include normative references will be developed later.

3 Definitions

For the purpose of this International Standard the following definitions apply.

3.1 Terms used to define the functions

3.1.1 whisking

to incorporate air and increase the volume of egg-whites

3.1.2 whipping cream

to incorporate air and increase the volume of cream

3.1.3 beating

to mix and incorporate air (e.g. cake batter)

3.1.4 heavy mixing

to incorporate ingredients evenly into a heavy mixture without chopping (e.g. fruit cake containing glacé cherries)

3.1.5 mixing

to combine ingredients evenly (e.g. pastry, pie)

3.1.6 kneading

to work a heavy mixture to form a smooth, pliable mass

3.1.7**blending/pureeing**

to make semi-solid ingredients liquid or to chop with liquid (e.g. babyfood, fruit, soup)

3.1.8**emulsifying**

to combine together insoluble liquids so that one or more is suspended in droplets throughout the other (e.g. mayonnaise)

3.1.9**chopping**

to reduce solid ingredients to small particles (e.g. meat, onions, almonds)

3.1.10**slicing**

to cut into slices (e.g. carrots, cucumber, leeks)

3.1.11**shredding**

to reduce to small strips (e.g. carrots, cheese)

3.1.12**french fry**

to cut potatoes into the shape of French fries

A1 3.1.13**juice separation**

to remove juice from fruit or vegetables by centrifugal force after breaking them up

3.1.14**citrus juice extraction**

to remove juice from citrus fruits by applying fruit halves to a rotating cone

3.1.15**coffee milling and grinding**

to break up whole coffee-beans in small particules by means of grinding discs, cylinders or cones and similar devices or by means of high-speed rotating cutting blades A1

3.2 Terms used to define the major families of appliances**3.2.1****mono-function appliance**

appliance designed to perform only one function

3.2.2**multi-function appliance**

appliance designed to perform more than one function

Appliances can be either hand-held type or table type.

NOTE Sometimes, for some appliances, it may be necessary to change the attachments or tools in order to perform the different functions.

A1 3.3 Terms used to define the convenience in use**3.3.1****splashing and spillage**

unintentional loss of ingredients during switching on and operation of an appliance

3.3.2**cleaning efficiency and cleaning time**

effort and time necessary to empty, clean and dry an appliance (or separate parts), including the disassembling of the appliance A1

4 List of measurements

This standard covers the following functions:

- whisking (Clause 6);
- whipping cream (Clause 7);
- beating (Clause 8);
- heavy mixing (Clause 9);
- mixing (Clause 10);
- kneading (Clause 11);
- blending/pureeing (Clause 12);
- emulsifying (Clause 13);
- chopping of meat (subclause 14.1);
- chopping of onions (subclause 14.2);
- chopping of almonds (subclause 14.3);
- slicing of carrots (subclause 15.1);
- slicing of cucumbers (subclause 15.2);
- slicing of leeks (subclause 15.3);
- shredding of carrots (subclause 16.1);
- shredding of cheese (subclause 16.2);
- french fries chipping (Clause 17);
- juice separation (Clause 18);
- citrus juice extraction (Clause 19);
- coffee milling and grinding (Clause 20).

This standard also covers:

-  — splashing and spillage (Clause 21);
- cleaning efficiency and cleaning time (Clause 22). 

5 General conditions for measurements

Unless otherwise specified, the measurements are conducted under the following conditions.

5.1 General

The attachments, speed, tools and quantities of ingredients used shall be those recommended by the manufacturer. In the absence of manufacturer's recommendations, the quantities quoted shall be used. The ingredients specified have been selected primarily to ensure uniform and reproducible results. A list of ingredients known to be suitable is given in Annex A. Unless otherwise stated, the appliance shall be operated in accordance with the manufacturer's instructions.

 All the tests from 6 to 20 have to be carried out three times (repeated twice). 

5.2 Electrical supply

The measurements shall be made at rated frequency and at a voltage which is within $\pm 0,5$ % of the rated voltage or the mean of the rated voltage range.

NOTE If the rated voltage differs from the nominal supply voltage of the country concerned, measurements carried out at rated voltage may be misleading. Therefore, for comparative testing the voltage used for the tests is to conform to the nominal supply voltage and this shall be reported.

5.3 Ambient temperature

The ambient temperature and the temperature of all utensils and ingredients shall be maintained at (20 ± 5) °C, unless otherwise specified.

NOTE In tropical climates, the ambient temperature may be (27 ± 5) °C but the temperature should be recorded.

5.4 Movement of hand-held machines

For whisking and whipping, a figure-of-eight movement shall be used unless it is impractical, in which case the beaters shall follow the shape of the bowl. The number of complete movements shall not exceed fifteen per minute.

For other operations, hand-held machines shall be moved as necessary to achieve good operation but at a speed not exceeding thirty movements per minute.

Hand-held machines supplied with bowl and stand shall be operated in accordance with the manufacturer's instructions. If no instructions are supplied and it is obvious that the bowl has to be moved, this is allowed but must be stated in the report.

5.5 Type of bowl

For appliances supplied with a bowl, the measurements shall be made using this bowl.

For appliances not supplied with a bowl, one of the bowls shown in Figure 1, Figure 2 and Figure 3 shall be used, choosing the most suitable for the function that has to be measured.

5.6 Ingredients

Where different ingredients have been used following particular national habits or particular country conditions a clear reference to these ingredients has to be given in the test report.

Where it is required that minimum and maximum quantities have to be processed, care has to be taken that the same proportion between all ingredients is maintained.

5.7 Preliminary tests

Sometimes and for some functions it may be necessary to carry out, for a given appliance or for a series of appliances, some preliminary tests in order to find out the time and the procedure to obtain the best result.

5.8 Baking temperature

Where in the standard it is required to bake the prepared mixture, baking has to be made in a normal (not forced convection) oven, at a suitable temperature in accordance with the oven manufacturer's instructions.

6 Whisking

6.1 Ingredients

Only fresh eggs shall be used according to Annex A.

The initial temperature of the eggs shall be $(23 \pm 2) ^\circ\text{C}$.

6.2 Procedure

Separate the egg-whites in a separate bowl, weigh individually, and use as required, (30 ± 2) g representing one egg-white. Put the egg-whites into the bowl, which is completely dry and free from grease. Measure and record the time taken to complete the whisking.

Whisking is considered to be achieved when the mixture is soft, glossy and moist in appearance and an impression made with a knife retains for at least 5 s. Tests are made to find the minimum and maximum number of eggs that can be satisfactorily whisked.

6.3 Specific volume

Immediately afterwards fill two Petri dishes (see Figure 4), each of known mass and volume, with a sample taken from the centre of the whole mixture of whisked egg-whites and weigh. Calculate the mean value of those two measurements.

The maximum volume is determined with four whisked egg-whites.

The specific volume shall be measured as described above. Pre-tests may be made to choose the whisking time suitable for the appliance to be tested.

For the time giving the highest specific volume the test has to be repeated. The mean value of the two measurements and the corresponding time are reported.

6.4 Result

Record the following:

- the minimum and maximum number of egg-whites that can be satisfactorily whisked;
- the maximum specific volume of the foam reached for four egg-whites and the time to reach this result;
- the attachments and setting used.

7 Whipping cream

7.1 Ingredients

Cream with no additives made from pasteurised milk and having a butter fat content of $(35 \pm 5) \%$ is used.

For the test either fresh cream or long-life cream can be used, but in the report it has to be stated what cream has been used.

An amount sufficient for one series of measurements shall be taken from the refrigerator where it has been stored for at least 24 h. The temperature of the cream shall be $(8 \pm 1) ^\circ\text{C}$.

7.2 Procedure

The ambient temperature and the temperature of all utensils is to be at $(20 \pm 5) ^\circ\text{C}$. The cream is put into the bowl and the whipping shall be started immediately. Whipping is considered to be achieved when an impression made with a knife is retained for at least 5 s. Tests are made to find the maximum quantity, in multiples of 100 g, that can be satisfactorily whipped.

7.3 Measurement of specific volume

Fill two Petri dishes (see Figure 4), each of known mass and volume, with a sample taken from the centre of the whole mixture of whipped cream and weigh. Calculate the mean value of those two measurements.

The maximum volume is determined with 200 g of whipped cream. The specific volume shall be measured as described above. Pre-tests may be made to choose the whipping time suitable for the appliance to be tested.

For the time giving the maximum specific volume the test is repeated. The mean value of the two measurements and the corresponding time are reported.

In order to determine any seepage one of the batches shall be put into a funnel (see Figure 6) and left at ambient temperature $(20 \pm 5) ^\circ\text{C}$ for 3 h.

7.4 Result

Record the following:

- the minimum and the maximum quantities that will give satisfactory results;
- the maximum specific volume of 200 g whipped cream and the time taken to reach this result;
- the attachments and setting used;
- the type of cream used (fresh or long-life);
- the amount of seepage.

8 Beating

8.1 Ingredients

The mass of ingredients to each egg is as follows:

- 50 g margarine;
- 50 g sugar;
- 50 g flour;
- 1 g baking powder or in accordance with the instructions of the manufacturer of the baking powder.

Initial temperature $(23 \pm 2) ^\circ\text{C}$.

8.2 Procedure

Tests of different amounts of ingredients are made to find the minimum and maximum quantities that can be satisfactorily beaten.

The result is satisfactory when all ingredients are evenly distributed.

Record the weight of the bowl and its attachments. Make a dough based on three eggs. Cut the margarine into cubes of approximately 20 mm. Place all the ingredients into the bowl and beat until satisfactorily mixed. Scrape down the sides of the bowl once, if necessary, with a spatula. Record the time needed for beating.

550 g of the dough is put into a long uncoated aluminium baking tin (see Figure 5). Place the tin in the middle of a preheated oven at a suitable temperature according to the oven manufacturer's instructions and bake until the cake is deemed to be done. The baking time and the temperature have to be recorded.

After carefully scraping the bowl and its attachments, weigh these and calculate the residual dough in the bowl and its attachments as a percentage of the total weight of the ingredients.

After the baking, the cake is taken out of the tin and turned upside down on a flat surface. It is covered with the tin and left for 24 h at room temperature. After the resting time the cake is weighed.

In order to determine the volume, the tin with the cake is filled up to the rim with mustard seeds. The seeds are measured in a measuring bowl. The difference between the known volume of the tin and the volume of the mustard seeds is the volume of the baked cake. If the cake has risen over the tin the measurement should be made, for instance, by aid of additional walls put on the top of the tin or by other suitable means, but without cutting the cake.

The cake has to be sliced evenly at different points (normally at 1/5, 2/5, 3/5 and 4/5 of its length).

The test has to be repeated and the mean value is calculated.

8.3 Result

Record the following:

- the minimum and maximum quantities that give satisfactory results, expressed as the number of eggs;
- the weight of residual dough in the working bowl and on its attachments;
- the baking time, the temperature and the type of the oven;
- the time taken to mix a dough of three eggs;
- volume of the baked cake;
- comments on the structure of the slices that has to be such that air pockets should be of even size and distribution;
- the attachments and setting used.

9 Heavy mixing

9.1 Ingredients

The mass of ingredients to each egg is as follows:

- 45 g margarine;
- 45 g sugar;
- 85 g flour;
- 3,5 g baking powder;
- 50 g currants, after soaking in tea;
- 50 g mixed dried fruit according to national availability.

Initial temperature (23 ± 2) °C.

9.2 Procedure

Tests of different amounts of ingredients are made to find the minimum and maximum quantities that can be satisfactorily mixed.

A satisfactory result is when all ingredients are evenly distributed and the fruits are not damaged.

Record the weight of the bowl and its attachments. Make a dough based on three eggs. Cut the margarine into cubes of approximately 20 mm. Place all the ingredients except the fruits into the bowl and process until satisfactorily mixed. At the end of the procedure the fruits are added and the appliance is operated till the moment when the fruits are distributed into the dough. Scrape down the sides of the bowl once, if necessary, with a spatula. Record the time needed for mixing.

950 g of the dough is put into a long uncoated aluminium baking tin (see Figure 5). Place the tin in the middle of a preheated oven at a suitable temperature according to the oven manufacturer's instructions and bake until the cake is deemed to be done. The baking time and the temperature have to be recorded.

After carefully scraping the bowl and its attachments, weigh these and calculate the residual dough on the working bowl and its attachments as a percentage of the total weight of the ingredients.

After baking the cake is taken out of the tin and turned upside down on a flat surface. It is covered with the tin and left for 24 h at room temperature. After the resting time the cake is weighed.

In order to determine the volume the tin with the cake is filled up to the rim with mustard seeds. The seeds are measured in a measuring bowl. The difference between the known volume of the tin and the volume of the mustard seeds is the volume of the baked cake. If the cake has risen over the tin the measurement should be made, for instance, by aid of additional walls put on the top of the tin or by other suitable means, but without cutting the cake.

The cake has to be sliced evenly at different points (normally at 1/5, 2/5, 3/5 and 4/5 of its length).

The test has to be repeated and the mean value is calculated.

9.3 Result

Record the following:

- the minimum and maximum quantities that give satisfactory results, expressed as the number of eggs;
- the weight of residual dough on the working bowl and its attachments;
- the baking time, the temperature and the type of the oven;
- the time taken to mix a dough of three eggs;
- volume of the baked cake;
- comments on the structure of the slices checking that fruits are evenly distributed into the slices and they are not damaged;
- the attachments and setting used.

10 Mixing

10.1 Ingredients

- 50 g margarine,
- 25 g water,

to each 100 g flour.

The temperature of the water shall be $(8 \pm 2) ^\circ\text{C}$; the other ingredients shall be at ambient temperature.

10.2 Procedure

Cut the margarine into cubes of approximately 15 mm and add them to the flour in the bowl.

Mix the ingredients until the mixture resembles fine crumbs of fresh bread; then add the water and process to form a satisfactory dough which can be removed from the working bowl easily.

Immediately roll the dough to a thickness of 4 mm, and put it into an uncoated aluminium pie tin (235 ± 5) mm diameter and 30 mm height lined with grease proof paper.

Put the tin into the refrigerator for 1 h. Line the bottom with greaseproof paper and fill the tin with dried beans. Then bake in a preheated oven at a suitable temperature according to the oven manufacturer's instructions until the pie crust is deemed to be done.

The pie crust has to be taken out of the oven at about 3/4 of baking time. Remove the paper and beans and return to the oven for the last quarter of the cooking time. The baking time and the temperature have to be recorded.

After the baking, the pie crust is removed from the oven, turned out upside down on a wire rack and left for 2 h at ambient temperature.

The test has to be repeated and the mean value is calculated.

10.3 Result

Record the following:

- maximum quantity that will give satisfactory mixing of the ingredients, expressed as the weight of flour in the recipe;
- time to mix the dough;
- an evaluation of whether the handling of the dough is easy and shows no imperfection;
- a visual evaluation (presence of fat spots) after baking;
- the attachments and setting used;
- the baking time, the temperature and type of the oven.

11 Kneading

11.1 Ingredients

Add to each 350 g flour:

- 5 g salt;
- 10 g margarine;
- 10 g sugar;
- 200 g water at a temperature of $(40 \pm 2) ^\circ\text{C}$ or according to yeast manufacturer's instructions;
- quick yeast (following the manufacturer's instructions).

11.2 Procedure

Place all the ingredients into the bowl and knead until satisfactory.

The dough is considered to be adequately kneaded when it loosens from the walls of the bowl, does not stick to the fingers and is pliant. Record the kneading time.

500 g dough is put into a long uncoated aluminium baking tin (see Figure 5). The tin has to be covered by a linen towel and it has to be placed for 45 min in a draught-free room at a temperature of $(23 \pm 2) ^\circ\text{C}$.

Place the tin in the middle of an oven, which has been preheated to a suitable temperature following the oven manufacturer's instructions.

After the baking, the bread is taken out of the tin and turned upside down on a flat surface. It is covered with the tin and left for 24 h at room temperature. After the resting time the bread is weighed.

In order to determine the volume, the tin containing the bread is filled up to the rim with mustard seeds.

The seeds are measured in a measuring bowl. The difference between the known volume of the tin and the volume of the mustard seeds is the volume of the baked bread. If the bread rises over the tin the measurement should be done for instance, by aid of additional walls put on top of the tin or by other suitable means but without cutting the bread.

The test has to be repeated and the mean value is calculated.

Tests of different amounts of ingredients are made to find the maximum quantity that can be satisfactorily kneaded.

11.3 Result

Record the following:

- the maximum amount of flour that can be adequately kneaded;
- the time taken to become a pliable mass that loosen from the walls and does not stick to the fingers;
- the weight of the loaf after baking;
- the volume of the loaf after baking;
- comments on the structure of the slices that has to be such that air pockets should be with similar dimensions, of even size and distribution;
- the attachments and setting used;
- the baking time, the temperature and type of the oven.

12 Blending/Pureeing

12.1 Ingredients

Carrots, potatoes, onions and parsley leaves.

12.2 Procedure

Peel and cut the uncooked carrots, potatoes and onions into regularly shaped pieces of about 10 g in weight. Make up the maximum quantity of soup by using multiples of 100 g (60 g vegetables and 40 g water).

Several batches are prepared having equal weights of carrots, potatoes and onions plus 0,5 g parsley leaves per batch.

The vegetables are blended and the resulting mass is passed through a 4 mm sieve. This operation may be aided by rinsing with cold water.

The test shall be repeated and the mean value recorded.

12.3 Result

Record the following:

- the maximum quantity in multiples of 100 g that will give a satisfactory result;
- the time taken to process;
- the mass of vegetables retained by the sieve expressed as a percentage of the original mass of vegetables;
- the attachments and setting used.

13 Emulsifying

13.1 Ingredients

Two minimum basic recipes have to be used and both have to be processed.

- | | |
|--------------------------|---------------------------|
| 1) Add to each egg yolk: | 2) Add to each whole egg: |
| — 15 g vinegar; | — 15 g vinegar; |
| — 250 g vegetable oil. | — 250 g vegetable oil. |

Before starting the test all ingredients shall be kept for at least 24 h at ambient temperature.

13.2 Procedure

Place the vinegar and egg in the working bowl and mix; then add the required quantity of oil at a rate of 100 g/min. Allow the mixture to stand for 10 min.

To judge the stability an amount of 100 g of mayonnaise has to be poured into a funnel (see Figure 6) and the seepage has to be weighed after 5 min, 10 min and 15 min.

The test has to be repeated and the mean value is calculated.

Record the following for each of the given recipes:

- NOTE** In case only one recipe has been processed this has to be clearly stated in the test report.

14.1 Chopping of meat

A cut of beef according to local availability and needs, trimmed of excess fat and gristle and cut into 20 mm cubes.

Temperature of the meat: $(8 \pm 2)^\circ\text{C}$.

Place in the bowl and process to desired fineness.

The test has to be repeated and the mean value is calculated.

Record the following:

- the minimum and maximum amounts that can be satisfactorily chopped, according to manufacturer's instructions;
- the time taken to chop;
- the regularity of and efficiency in chopping;
- the attachments and setting used.

14.2.1 *Ingredients*

Use onions of approximately the same shape and size.

Peel and cut the onions into pieces regularly shaped of about 10 g in weight.

Put the onions in the bowl. Separate tests shall be made until the onions are deemed to be:

- a) *coarsely chopped*;
b) *finely chopped*.

Scrape the sides of the bowl down once with a spatula, if necessary.

The test has to be repeated and the mean value is calculated.

14.2.3 Result

Record the following:

- the minimum and maximum quantities that can be satisfactorily chopped;
- the time taken to chop:
 - coarse;
 - fine;
- an evaluation of the chopped result made up of the regularity and clean cut of the result, the amount of moisture and bruising;
- the attachments and setting used.

NOTE In order to facilitate the evaluation of the test result, the batches processed by the appliance can be compared with other batches chopped by hand.

14.3 Chopping of almonds

14.3.1 Ingredients

Whole shelled almonds with skin.

14.3.2 Procedure

Put the almonds in the bowl or feed the appliance as specified by the manufacturer's instructions and chop, until they are:

- a) *coarsely chopped;*
- b) *finely chopped;*
- c) *extra finely chopped (as a powder).*

After each processing the mixture is put back in the bowl for the following process, until the required result is reached.

The test has to be repeated and the mean value is calculated.

14.3.3 Result

Record the following:

- the minimum and maximum useful amounts that can be satisfactorily chopped;
- the time taken to chop;
- an evaluation of the chopped result: regularity and clean cut of chopping (mostly for a) and the quantity of unprocessed almonds remaining in a sieve of:
 - 4 mm for a)
 - 2,5 mm for b)
 - 0,8 mm for c)

expressed as a percentage of the initial amount;

- the attachments and setting used.

15 Slicing

15.1 Slicing of carrots

15.1.1 Ingredients

500 g prepared carrots; diameter 15 mm to 25 mm; length 100 mm to 150 mm.

The carrots shall be cut to approximatively the same length and in such a way that they do not protrude from the rim of the feeder.

15.1.2 Procedure

The test is carried out with each slicing disc, if any. Feed the carrots through with the pusher and then operate the appliance. Between each feeding the appliance has to be stopped.

With bowls having a small capacity, it may be necessary to process more than one batch.

In which case, record:

- *the total time taken;*
- *the time to empty the bowl.*

The maximum number of carrots that can be contained in the feeder is introduced for each feeding.

The test has to be repeated and the mean value is calculated.

15.1.3 Result

Record the following for each disc:

- the weight of the undamaged slices, the weight of damaged slices and weight of residue;
- the regularity of and occurrence of defects in the shape of the slices;
- the setting used;
- the measured thickness of the cutting opening;
- the total time taken to slice the whole quantity of carrots.

15.2 Slicing of cucumber

15.2.1 Ingredients

One unpeeled greenhouse cucumber of known weight and diameter which fits entirely into the feeding tube and has a length greater than the feeding tube.

NOTE Where the feeding tube is too small, trim the side of the cucumber to fit, preserving the maximum amount of peel.

15.2.2 Procedure

The test is carried out with each slicing disc, if any. Feed the cucumber through with the pusher and then operate the appliance. With bowls having a small capacity it may be necessary to process in more than one batch. In which case record:

- *the total time taken;*
- *the time to empty the bowl.*

The test has to be repeated and the mean value is calculated.

15.2.3 Result

Record the following for each disc:

- the weight of the undamaged slices, the weight of damaged slices and weight of residue;
- the regularity of the slices and the occurrence of defects;
- how cleanly the slices are cut;
- the setting used;
- the measured thickness of the cutting opening;
- the total time taken to slice the quantity.

15.3 Slicing of leeks

15.3.1 Ingredients

500 g prepared leeks, cut at the part where they start opening and after having taken off the damaged outside leaves and roots.

The leeks can be washed, if necessary, on the outside only.

The dimension of the leek may be larger than the appliance can accept. In this case remove the outside leaves to reduce the diameter.

15.3.2 Procedure

The test is carried out with each slicing disc, if any. Feed the leeks through with the pusher and then operate the appliance. Between each feeding the appliance has to be stopped. With bowls having a small capacity it may be necessary to process in more than one batch. In which case record:

- the total time taken;*
- the time needed to empty the bowl;*

The maximum number of leeks that can be contained in the feeder is introduced for each feeding.

The test has to be repeated and the mean value is calculated.

15.3.3 Result

Record the following for each disc:

- the weight of the undamaged slices, the weight of damaged slices and weight of residue;
- the regularity of and occurrence of defects in the shape of the slices;
- how cleanly the slices are cut;
- the setting used;
- the measured thickness of the cutting opening;
- the total time taken to slice the whole quantity of leeks.

16 Shredding

16.1 Shredding of carrots

16.1.1 Ingredients

500 g prepared carrots, diameter 20 mm to 40 mm.

16.1.2 Procedure

The test is carried out with each shredding disc, if any. Cut the carrots of a sufficient length to be placed horizontally into the feeder stacked to the rim.

Push the carrots with the pusher and operate the appliance. Between each feeding the appliance has to be stopped. With bowls having a small capacity it may be necessary to process in more than one batch. In which case record:

- the total time taken;*
- the time needed to empty the bowl.*

The maximum number of carrots that can be contained in the feeder is introduced for each feeding.

The test has to be repeated and the mean value is calculated.

16.1.3 Result

Record the following for each disc:

- the weight of the undamaged shreds, the weight of the damaged shreds and the weight of residue;
- the regularity of the shreds; whether they are bruised and whether there is visible moisture loss;
- the setting used;
- the measured thickness of the cutting opening;
- the total time taken to shred the whole quantity of carrots.

16.2 Shredding of cheese

16.2.1 Ingredients

200 g of hard cheese. Temperature of the cheese $(8 \pm 2) ^\circ\text{C}$.

NOTE Cheddar or Emmenthal or Gouda cheese or cheese according to local availability may be used for the test.

16.2.2 Procedure

The test is carried out with each shredding disc, if any. Cut the cheese over a length and with a size suitable to be placed horizontally into the feeder stacked to the rim.

Push the cheese with the pusher and operate the appliance. Between each feeding the appliance has to be stopped.

With bowls having a small capacity it may be necessary to process in more than one batch. In which case record:

- *the total time taken;*
- *the time needed to empty the bowl.*

The maximum quantity of cheese that can be contained in the feeder is introduced for each feeding.

The test has to be repeated and the mean value is calculated.

16.2.3 Result

Record the following for each disc:

- the weight of the undamaged shreds, the weight of the damaged shreds and the weight of the residue;
- the regularity of the shreds; whether they are sticky;
- the setting used;
- the measured thickness of the cutting opening;
- the type and fat content of the cheese used for the test;
- if the disc is covered by an uniform squashed layer of cheese;
- the total time taken to shred the whole quantity of cheese.

17 French fries chipping

17.1 Ingredients

500 g of prepared potatoes.

17.2 Procedure

Use the special discs for French fries chipping. With each disc, cut the potatoes length-ways to a suitable size to fit the feeding tube, and stack the pieces horizontally to the rim.

Feed the potatoes through with the pusher and then operate the appliance. Between each feeding the appliance has to be stopped. With bowls having a small capacity it may be necessary to process in more than one batch. In which case record:

- *the total time taken;*
- *the time needed to empty the bowl.*

The maximum number of potatoes that can be contained in the feeder is introduced for each feeding.

The test has to be repeated and the mean value is calculated.

17.3 Result

Record the following for each disc:

- the weight of the undamaged pieces, the weight of the damaged pieces and the weight of the residues;
- the regularity of the chips; whether they are bruised and whether there is visible moisture loss;
- the setting used;
- the measured thickness of the cutting opening;
- the total time taken to chip the whole quantity of potatoes.

18 Juice separation

A1 18.1 Ingredients

- Carrots with a diameter of 20 mm to 40 mm, having been soaked for 24 h in water, topped and tailed, cleaned, wiped dry and cut to pieces of approximately 20 mm in length.
- Apples washed, dried, uncored and cut into pieces for inserting into the feeding tube of the appliance.
- Grapes, washed, dried and freed of stems.
- Firm tomatoes, washed, dried and cut into pieces for inserting into the feeding tube of the appliance.

18.2 Procedure

18.2.1 Separation of 350 g of fruit or vegetables

Process 350 g of fruit or vegetables. Allow the appliance to run for 10 s after the last piece of fruit or vegetable has been processed and then record the processing time. Allow the machine to stand for 60 s after switching off. Then record the quantity of unstrained juice in grams. Pour the juice through a 0,25 mm sieve. For tomatoes use a sieve of 1,4 mm. Wait 60 s after completing the filtering process and record the amount of strained juice in grams.

Clean the appliance.

The test has then to be repeated.

18.2.2 Separation of the maximum quantity of fruit or vegetables

Process the maximum quantity of fruit or vegetables possible until the machine malfunctions or overflows or requires emptying. Record the quantity of fruit or vegetables processed. In the case of apples, the test must be carried out to determine the point at which the appliance becomes clogged.

Repeat the test, reducing the amount of fruit or vegetables to the next lowest multiple of 50 g.

Cleaning time between subsequent batches: record the time taken to dismantle the appliance, to remove the pulp and reassemble, without rinsing the appliance.

18.3 Results

Record the following for each type of fruit or vegetable:

- the type of fruit or vegetable;
- the mean value of the unstrained juice as a percentage of the value of the weight of fruit or vegetable used (for the test of **18.2.1**);
- the mean value of the strained juice as a percentage of the value of the weight of fruit or vegetable used (for the test of **18.2.1**);
- the mean value of the time taken to process 350 g of fruit or vegetables (for the test of **18.2.1**);
- the maximum quantity of fruit or vegetables which the appliance can accommodate without malfunctioning;
- the time needed to process the maximum quantity of fruit or vegetables;
- the cleaning times between subsequent batches;
- the reason which limits the maximum quantity (malfunctioning or overflowing or emptying);
- the attachment and setting used.

19 Citrus juice extraction

19.1 Ingredients

- Lemons
- Oranges
- Grapefruit

Fruit which is known to be suitable for juice extraction should be selected. **A1**

A1 19.2 Procedure

Weigh five lemons, five oranges and four grapefruit. For each type of fruit the weight shall be recorded as mass A.

The citrus fruit shall, as far as possible, be equally halved. Where a choice of reaming cones is available, the most appropriate for each fruit shall be selected.

Sufficient downward pressure should be applied by hand to extract as much juice and flesh as possible. If the appliance cannot accommodate this amount of fruit in one operation, record the number of fruit halves which can be accommodated and state whether the juice-collecting vessel or the strainer is the reason. After the last fruit half has been reamed, allow the juice to percolate from the strainer for 2 min and then record the time.

Weigh the juice and pulp that have passed through the strainer as mass B and the skins as mass D and record both values.

Strain the juice and pulp through a 1,4 mm sieve. The juice and pulp in the sieve may be agitated gently during this process to extract as much of the juice as possible, but care should be taken not to squeeze the pulp through the sieve. Record the mass of the strained juice as mass C. Repeat the test. Clean the appliance in between.

If any part of the pith of the outer skin becomes pierced, the extraction is not considered satisfactory.

19.3 Extractor efficiency

Calculate the extractor efficiency for each test from the formula:

$$\frac{B}{A} \times 100 \%$$

19.4 Strainer efficiency

Calculate the strainer efficiency for each test from the formula:

$$\frac{C}{B} \times 100 \%$$

19.5 Results

Record the following for each type of fruit:

- the type of fruit;
- the total weight of the whole fruit for each test;
- extractor efficiency as a mean value of the measurements;
- strainer efficiency as a mean value of the measurements;
- time to process;
- the attachments and setting used;
- whether the appliance is unable to accommodate the specified amount of fruit;
- the weight of unprocessed fruit remaining.

20 Coffee milling and grinding**20.1 Ingredients**

Unbroken coffee-beans kept for 24 h on an open tray, one layer thick, at room temperature.

20.2 Procedure

Two quantities should be tested: 20 g and the maximum quantity recommended by the manufacturer; if no recommendation is given, the maximum quantity is 100 g. For each quantity, three determinations should be made: coarse, medium and fine. For each determination, the test has to be repeated and the mean value calculated. **A1**

A) *The appliance is operated with the setting or the time recommended by the manufacturer for each test. After each test the whole quantity is sieved using 200 mm diameter ISO sieves (see ISO 3310-1)¹⁾. The sieves are operated by means of an Endecott test sieve shaker having:*

<i>horizontal amplitude:</i>	<i>10 mm at a frequency of 1 440 vibrations per minute;</i>
<i>vertical amplitude:</i>	<i>0,8 mm at a frequency of 300 vibrations per minute.</i>

The result is judged by the definitions of coarse, medium and fine as follows:

- a) *“COARSE” more than 50 % of the grounds are larger than 0,71 mm;*
- b) *“MEDIUM” more than 50 % of the grounds are larger than 0,355 mm, but smaller than 0,71 mm;*
- c) *“FINE” more than 50 % of the grounds are smaller than 0,355 mm.*

20.3 Results

Record the following:

- the setting used and total processing time;
- percentage of coffee grains retained by each size of sieve;
- percentage of coffee grains passed through the smallest sieve;
- for milling, the maximum mass in grams milled in one batch;
- for grinding, the rate of output in grams per minute and the maximum quantity that can be ground in one batch.

21 Splashing and spillage

21.1 Ingredients

No ingredients other than those mentioned in Clauses 6 to 20 are to be used.

21.2 Procedure

During each test of Clauses 6 to 20 observe and record any splashing and spillage, noting when, nature and degree. Take into account an appropriate working surface, since this is mainly a visual judgement.

21.3 Result

The amount of splashing/spillage is classified as follows: none; slight; excessive.

22 Cleaning efficiency and cleaning time

22.1 Void.

22.2 Procedure

*Following each of the tests carried out, remove as much of the processed ingredients and residue as possible using the utensil provided or recommended by the manufacturer or any suitable utensil in case the manufacturer does not provide information. **A)***

¹⁾ ISO 3310-1:1990, *Test sieves — Technical requirements and testing — Part 1: Test sieves of metal wire cloth.*

A1) After that, a visual inspection should be made to check the level of difficulty for cleaning the appliance, taking into consideration aspects such as (listed with decreasing importance):

- dismantling (record the time);
- risk of cutting;
- sharp edges;
- food-trapping areas;
- number of single parts to be cleaned;
- dishwasher resistance;
- easy cleaning by hand; with or without water; with or without detergent (record the time);
- provision of special cleaning utensils provided by the manufacturer;
- discoloration of parts during operation.

22.3 Results

Record and specify the number of parts necessary to clean, following the manufacturer's instructions.

Record if the appliances or parts are dishwasher-resistant.

Record time for dismantling and for cleaning by hand.

Record the ease of cleaning.

The ease of cleaning for each function is classified as follows: very good; satisfactory; poor. **A1**

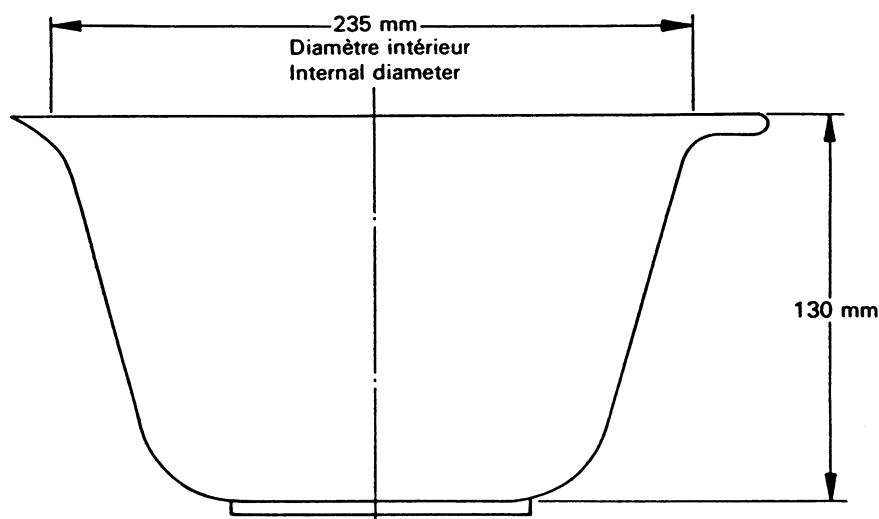
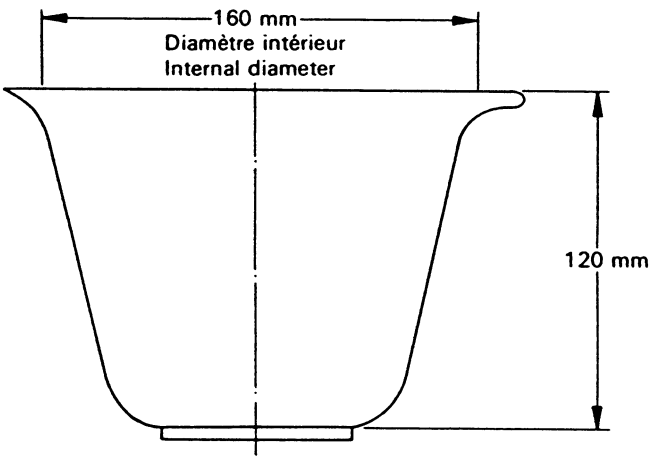


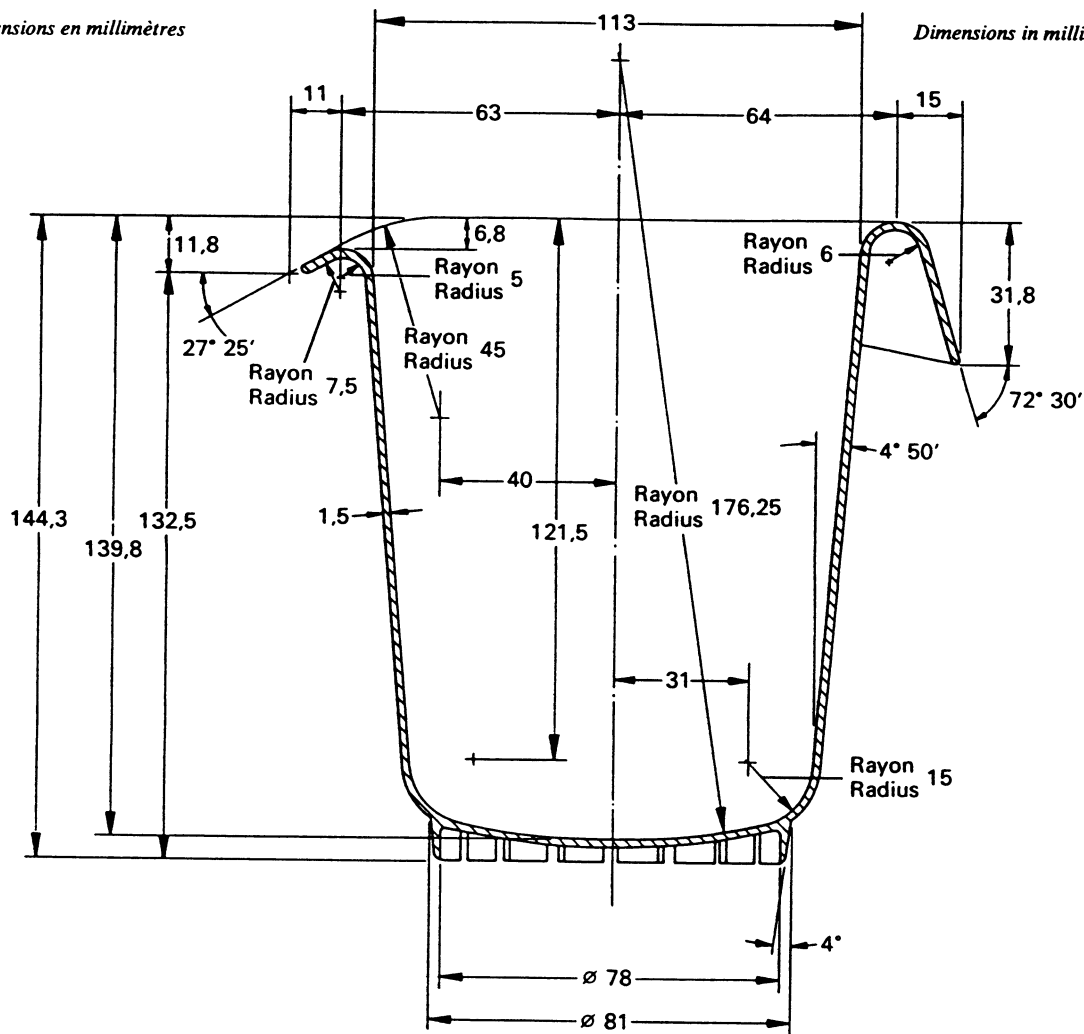
Figure 1 — Standard 4 I “MEPAL” mixing bowl



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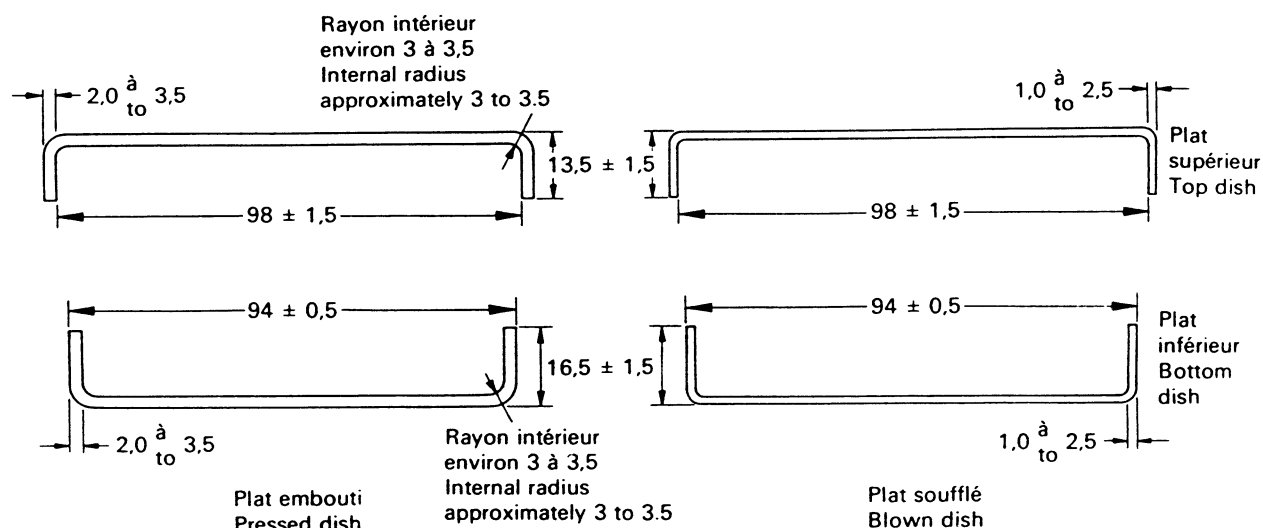
Figure 2 — Standard 1,5 I “MEPAL” mixing bowl

Dimensions en millimètres



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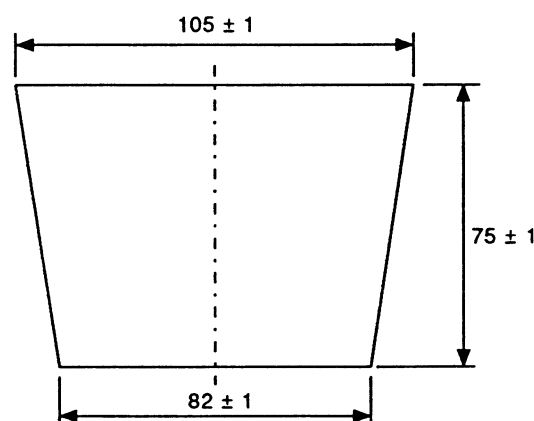
Figure 3 — Standard 1 I mixing bowl for use with immersion blenders



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*Dimensions en millimètres**Dimensions in millimetres*

- 1) The dish may be pressed or blown: glass or steel sheet.
- 2) The dish shall be circular.
- 3) Edges of top et bottom dishes shall be ground flat or moulded parallel with the base.
- 4) The top dish shall fit loosely on the bottom dish without rocking.
- 5) The base of both top and bottom dishes shall be approximately a plane surface. The material thickness shall be as uniform as possible so that the insides of both top and bottom dishes shall be approximately a plane surface.

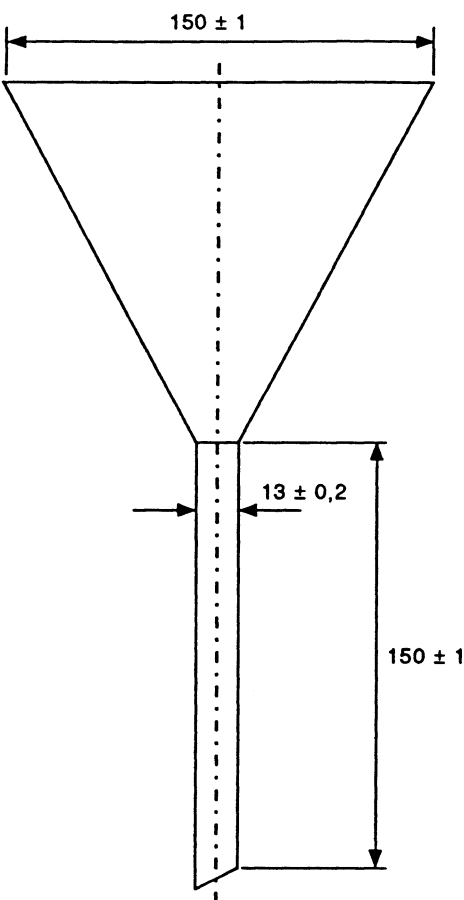
Figure 4 — Dish for measurement of specific volume

CEI-IEC 091193

*Dimensions en millimètres**Dimensions in millimetres*

The tin has to be as long as necessary for the quantity of batter under test and with the angles on the long side the same as in the figure. Fill the tin half full with batter

Figure 5 — Baking tin



CEI-IEC 092/93

NOTE The values for the diameters of the glass funnel are relevant to internal diameters.

Figure 6 — Glass funnel

National annex NA (informative)

Committees responsible

The United Kingdom participation in the preparation of this European Standard was entrusted by the Light Electrical Engineering Standards Policy Committee (LEL/-) to Technical Committee LEL/59 upon which the following bodies were represented:

- Association of Manufacturers of Domestic Electrical Appliances
- Consumer Policy Committee of BSI
- Consumers' Association
- Department of Trade and Industry (Electricity Division)
- Electricity Association
- Portable Electric Tool Manufacturers' Association
- Rotating Electrical Machines Association (BEAMA Ltd.)

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